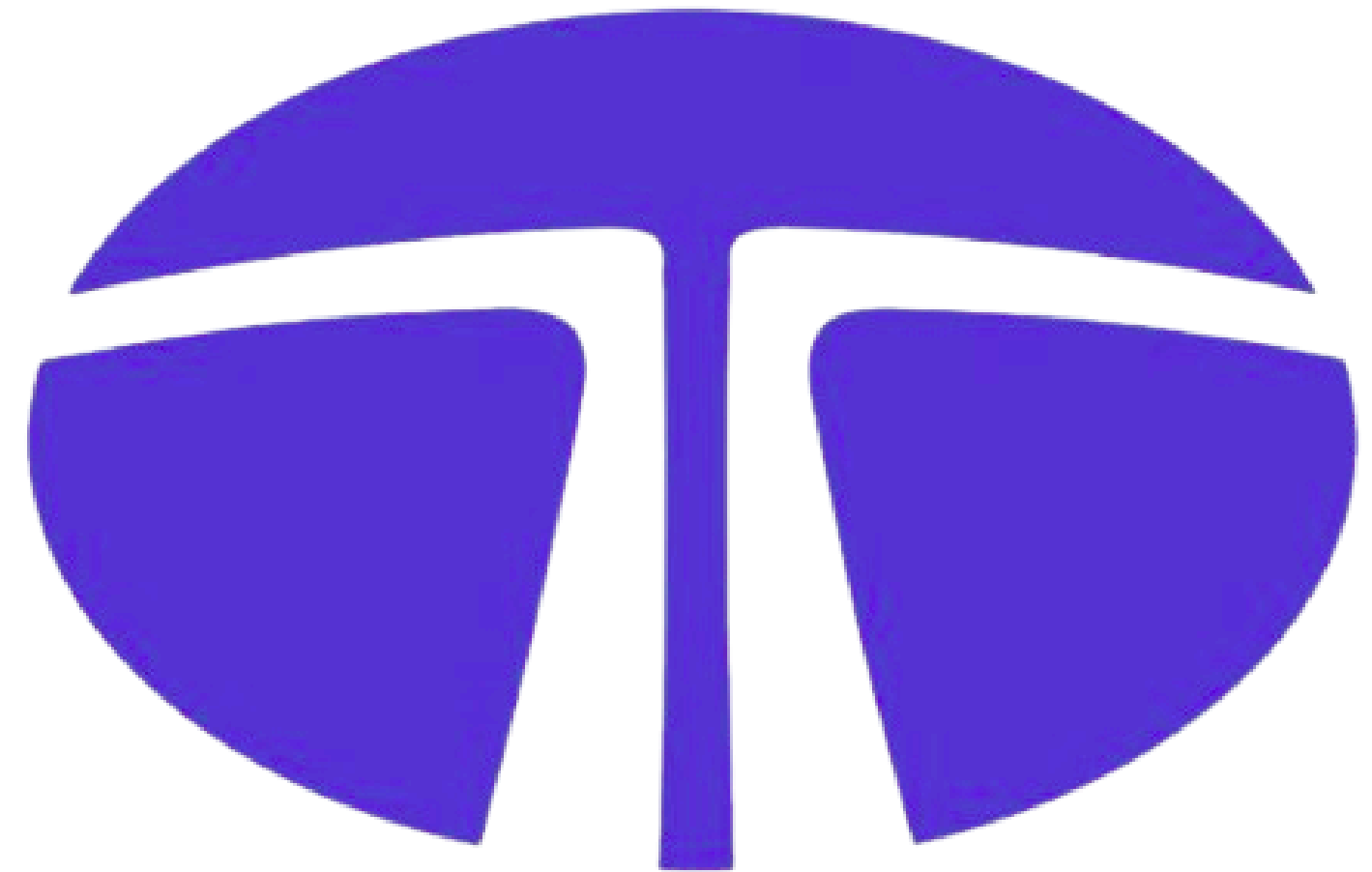


TATA Motors Sales Intelligence Dashboard

Full Stack Data Analysis | Python • SQL
• Power BI • Streamlit

- **PRESENTED BY**
Kushang Bhattacharya
- **DEPARTMENT**
Data Analytics



TATA MOTORS

**Dataset: Synthetic data for portfolio demonstration purposes. All monetary values represented in USD.*



ABOUT TATA MOTORS

1. Basic Profile

- Established: 1945
- Founder: J.R.D. Tata
- Headquarters: Mumbai, India
- Parent Group: Tata Group

2. FY25 Financials (Latest 2025 Data)

- Consolidated Revenue: ₹4,39,695 Crore
- Net Profit (PAT): ₹28,149 Crore
- Status: Turned net auto-cash positive

3. Core Business (What They Do)

- Passenger Cars: Making high-safety SUVs and cars (Nexon, Punch, Harrier).
- Commercial Vehicles: Market leader in trucks, buses, and logistics vans.
- Electric Vehicles: Dominates India's EV market with a 70%+ share.
- Luxury Segment: Proud owner of global iconic brands Jaguar Land Rover (JLR).

4. Key Achievements

- Safety Pioneer: First Indian automaker to get a 5-Star Global NCAP safety rating.
- Global Reach: Active operations across 125+ countries with R&D centers in India, UK, and South Korea

Work Flow

01



Python
Data Cleaning
Exploratory Data Analysis
Tool: Jupyter Notebook

02



SQL
MySQL Connection
Business Queries
Tool: MySQL Workbench

03



Power BI
5 Page Executive
Dashboard
Tool: Power BI Desktop

04



Streamlit
Live Interactive Web App
Tool: VS Code



Data Cleaning — Python (Jupyter Notebook)

Data Cleaning — Python

Step 1: Data Loading

- Loaded raw CSV — **250,000** rows, **34** columns
- Checked info, null values, first 10 rows

Step 2: Null Value Treatment

- promotion nulls → filled with "No Promotion"
- First time buyers → previous_car_brand & previous_car_model = "Not Applicable"
- Remaining nulls → filled with "Unknown"

Step 3: Text Standardization

- Removed extra spaces from all text columns
- Standardized all text to Title Case

Step 4: Data Validation

- Checked negative values in average_sale_price
- Checked negative values in dealer_discount_pct

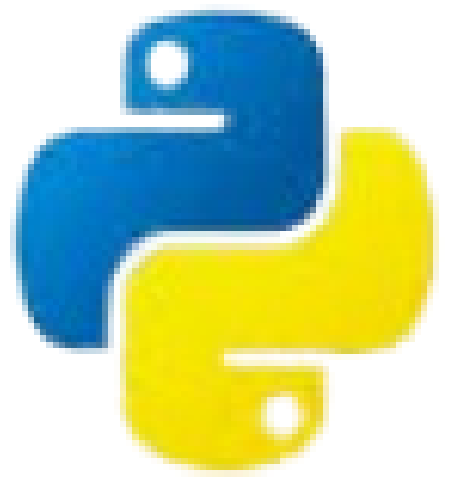
Step 5: Export

- Saved as TATA_car_sales_cleaned.csv
- Final Result: **Zero Null Values**

Exploratory Data

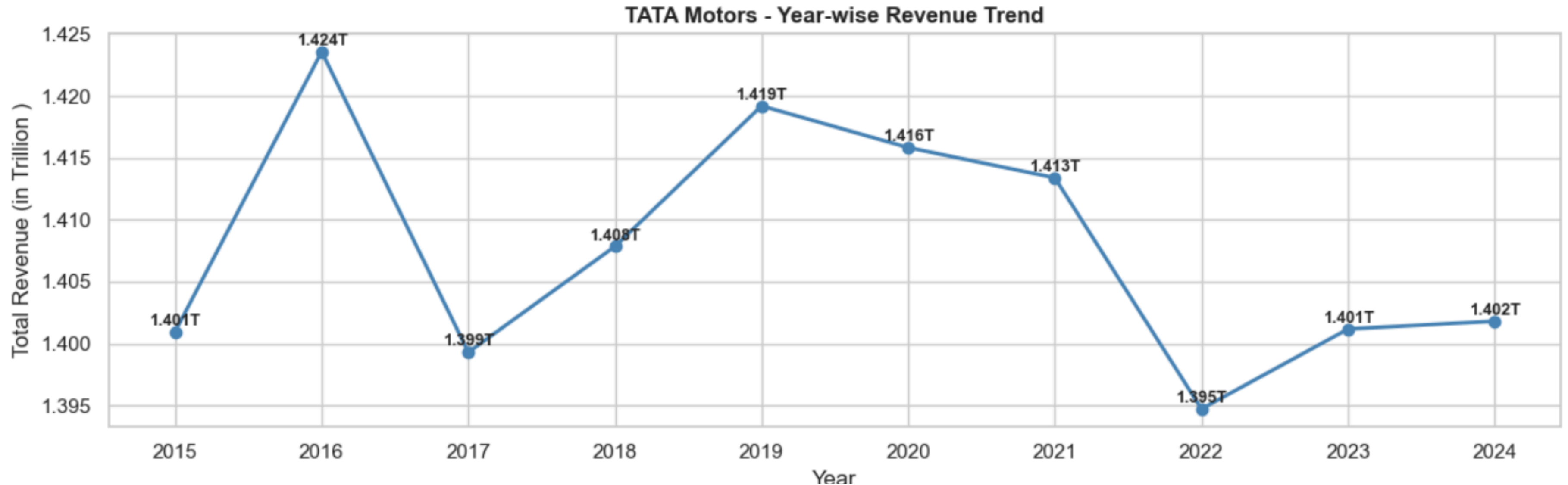


Analysis — Python



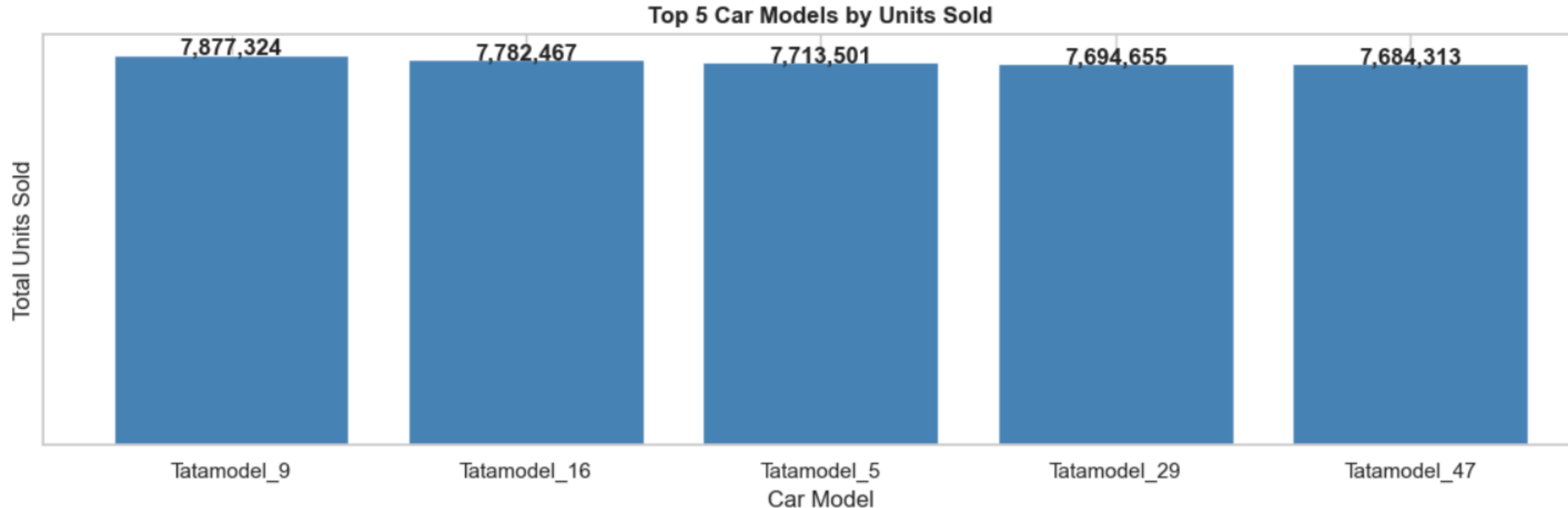
(Jupyter Notebook)

Year-wise Revenue Trend



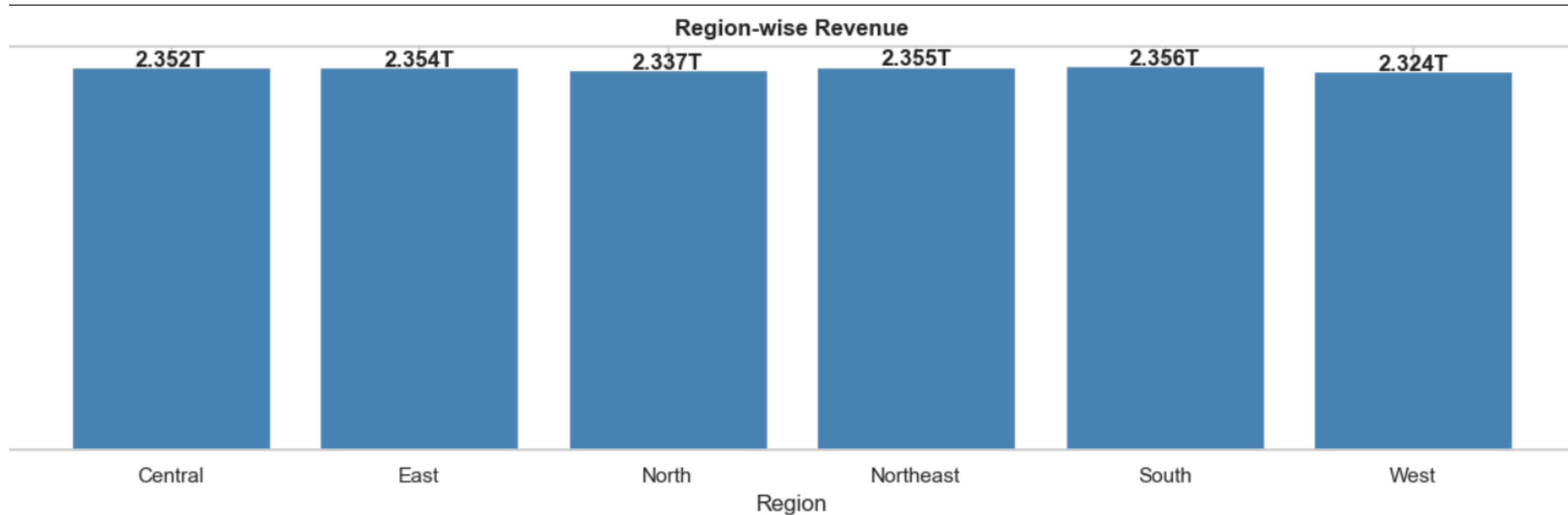
- Revenue stable between 1.395T – 1.424T
- Peak: 2016 (1.424T)
- Lowest: 2022 (1.395T)
- Overall: No Dramatic Growth or Decline

Top 5 Car Models by Units Sold



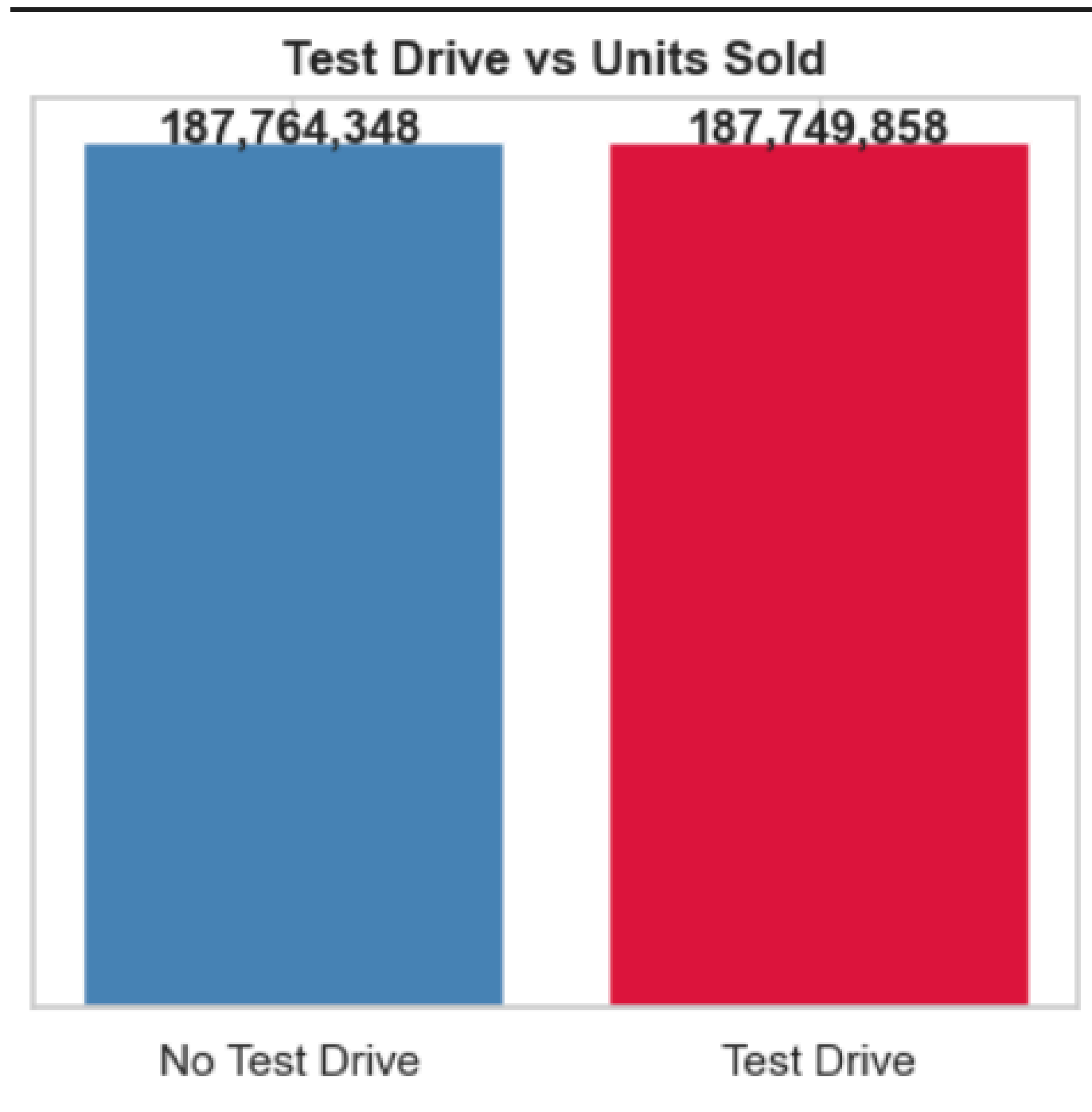
- Tatamodel_9 – Highest (7,877,324 units)
- Tatamodel_16 – 2nd Highest (7,782,467 units)
- All top 5 models closely competitive
- Difference between 1st & 5th = ~193,011 units

Region-wise Revenue



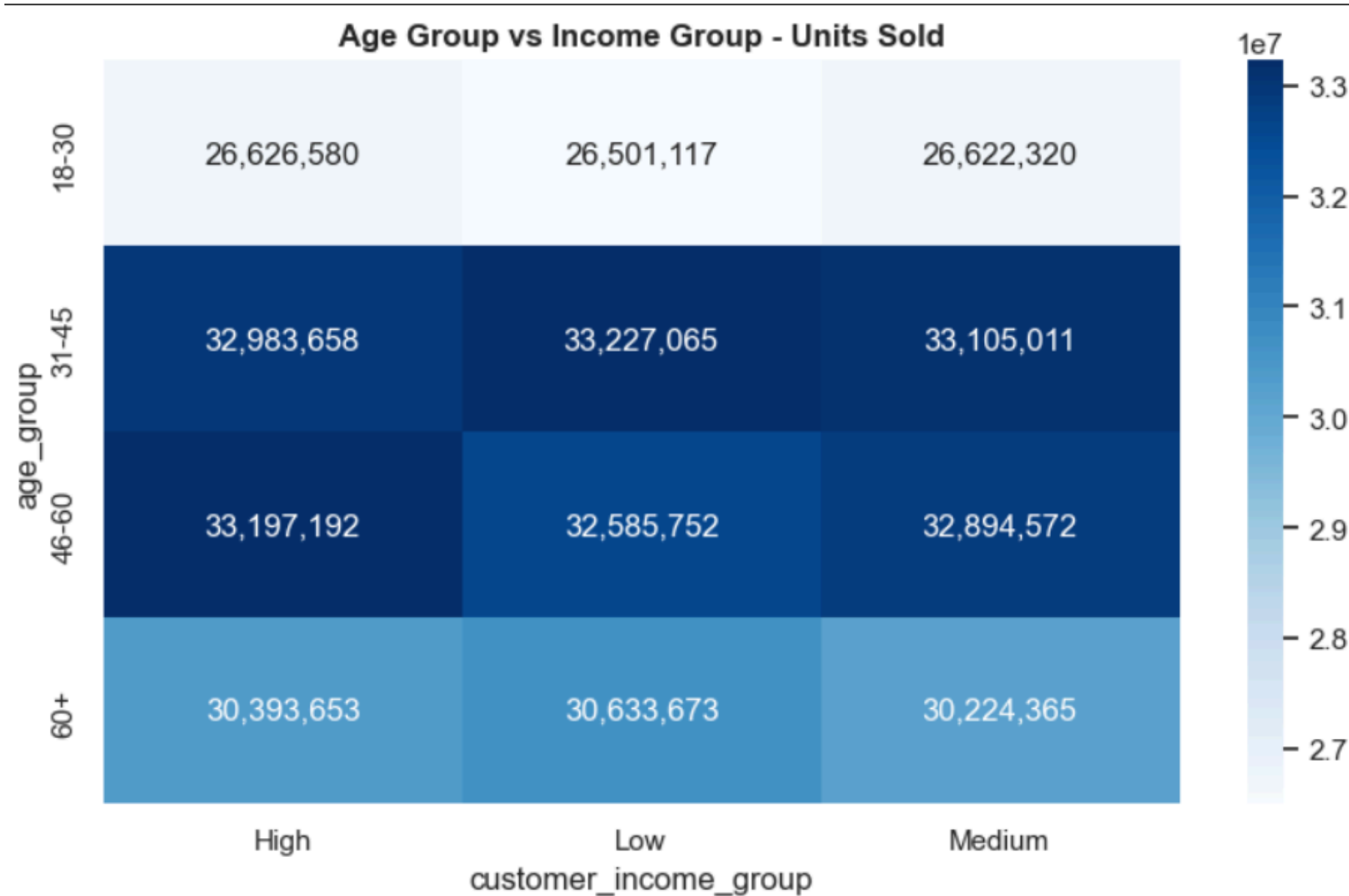
- South — Highest (2.356T)
- West — Lowest (2.324T)
- All 6 regions almost equal revenue
- No single dominant region found

Test Drive vs Units Sold



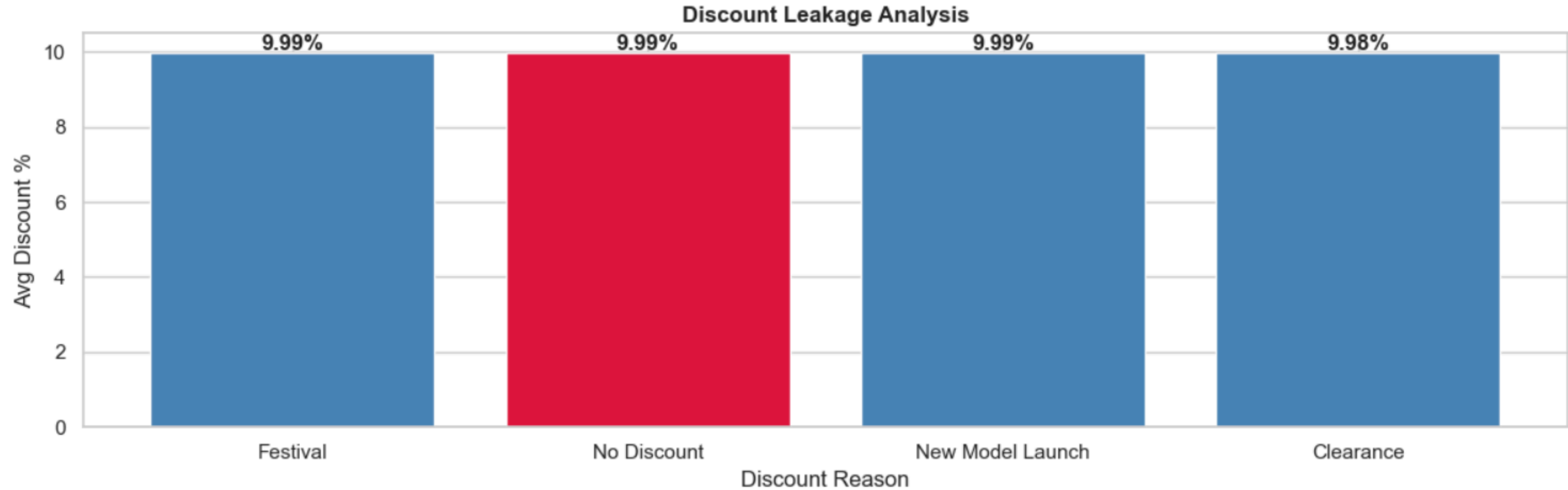
- No Test Drive: 187,764,348 units
- Test Drive: 187,749,858 units
- Difference: Negligible
- Insight: Test drive has no impact on sales

Age Group vs Income Group



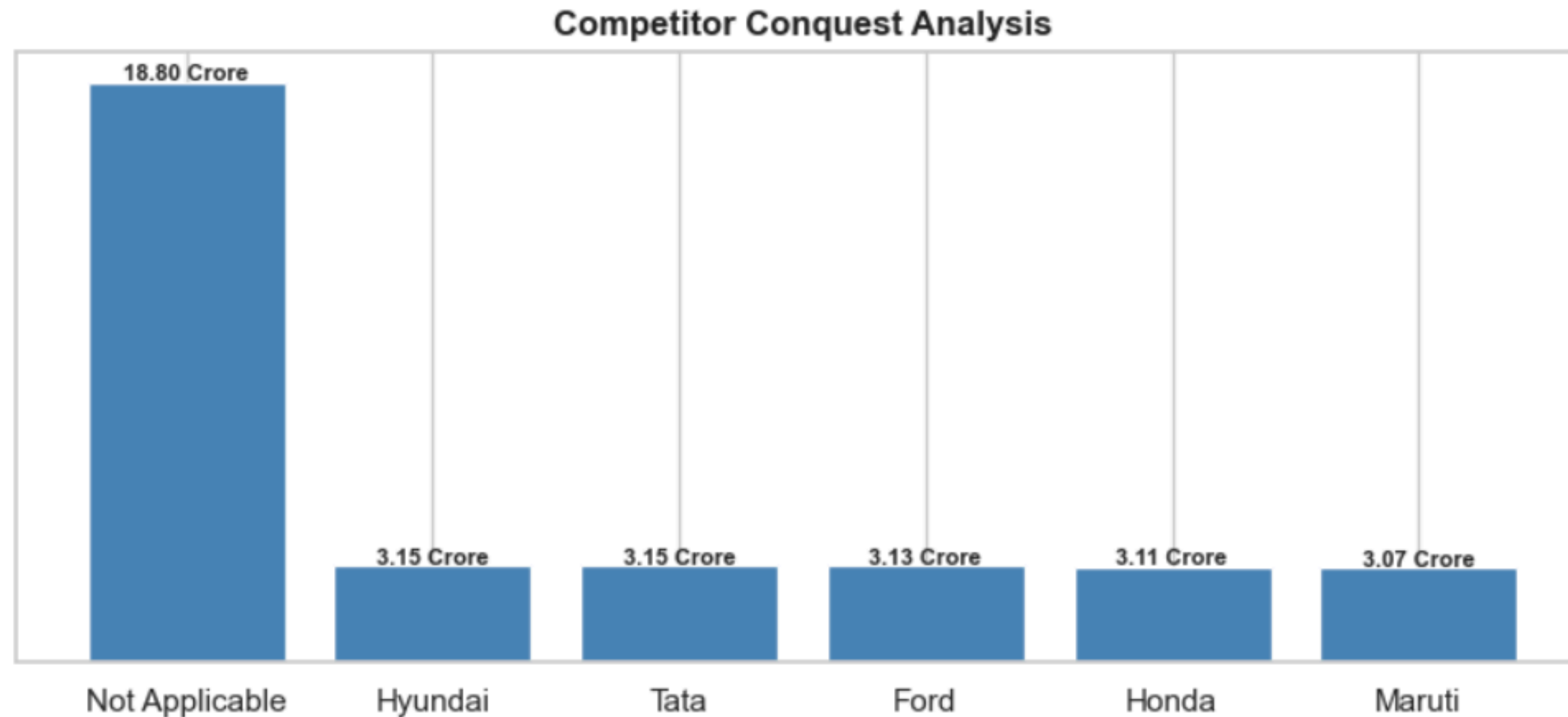
- 31-45 age group = highest buying segment
- 46-60 High Income = 33,197,192 units (peak)
- 18-30 = lowest buying segment
- All income groups buy equally across ages

Discount Leakage



- All 4 reasons show ~9.99% avg discount
- "No Discount" transactions = 9.99% discount
- Clear discount leakage identified (red bar)
- Financial loss without valid reason

Competitor Conquest



- **Not Applicable (First Time Buyers) = 18.80 Cr**
- **Hyundai conquest = 3.15 Cr units**
- **Tata loyal customers = 3.15 Cr units**
- **Maruti conquest = lowest (3.07 Cr)**
- **Opportunity: Target Maruti customers more**

Delivery Time vs Customer Rating



- Rating range: 3.22 – 3.27 across all days
- No correlation between delivery & rating
- Customers rate based on product, not delivery



SQL Business Queries — MySQL

Top Countries by Avg Sale Price

```
3  -- Query 1: Top Countries by Average Sale Price
4
5  • SELECT country,
6      ROUND(AVG(average_sale_price), 2) AS avg_price
7  FROM tata_sales
8  GROUP BY country
9  ORDER BY avg_price DESC;
```

- India = Highest avg price (\$37,649)
- Japan = Lowest avg price (\$37,362)
- Insight: All countries uniform pricing
- No premium market identified

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	country	avg_price			
▶	India	37649.49			
	Uk	37635.05			
	Usa	37597.67			
	Germany	37476.12			
	Australia	37446.74			
	France	37405.87			
	Japan	37362.6			

Discount Leakage

```
11  -- Query 2 : Discount Leakage Analysis
12
13  •  SELECT discount_reason,
14         ROUND(AVG(dealer_discount_pct),2) AS discount_percentage
15  FROM tata_sales
16  GROUP BY discount_reason
17  ORDER BY discount_percentage DESC;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	discount_reason	discount_percentage			
▶	New Model Launch	9.99			
	No Discount	9.99			
	Festival	9.99			
	Clearance	9.98			

- New Model Launch = 9.99% avg discount
- No Discount = 9.99% avg discount
- Festival = 9.99%
- Clearance = 9.98%
- Insight: "No Discount" transactions still showing ~10% discount — Clear leakage!

First Time Buyer % by Region

```
18  -- Query 3 First-time buyer % by region
19
20  • SELECT region,
21      ROUND(SUM(CASE WHEN is_first_time_buyer = 'True' THEN 1 ELSE 'False' END) * 100.0 / COUNT(*), 2) AS first_time_buyer_pct
22  FROM tata_sales
23  GROUP BY region
24  ORDER BY first_time_buyer_pct DESC;
25
26
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
region	first_time_buyer_pct		
West	50.47		
East	50.31		
Central	49.88		
Northeast	49.85		
South	49.72		
North	49.48		

- West = Highest (50.47%)
- East = 50.31%
- North = Lowest (49.48%)
- Insight: West & East = strongest conquest markets for TATA

Sales Channel Performance

```
27  -- Query 4 Sales Channel Performance
28
29  •  SELECT sales_channel,
30         SUM(units_sold) AS total_sold,
31         ROUND(AVG(dealer_discount_pct), 2) AS discount_percentage,
32         ROUND(AVG(customer_rating), 2) AS customer_ratings
33  FROM tata_sales
34  GROUP BY sales_channel
35  ORDER BY total_sold DESC;
```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 				
	sales_channel	total_sold	discount_percentage	customer_ratings
▶	Online	94222810	10	3.25
	Fleet	93858905	9.96	3.24
	Dealership	93787855	9.98	3.25
	Direct	93644636	10	3.25

- Online = Best (94,222,810 units)
- Fleet = 93,858,905 units
- Dealership = 93,787,855 units
- Direct = Lowest (93,644,636 units)
- Insight: Digital strategy working well

Year over Year Growth

```
37  -- Query 5  Year over Year Growth Rate
38
39  • WITH yearly AS (
40      SELECT year,
41             SUM(units_sold * average_sale_price) AS total_revenue
42      FROM tata_sales
43      GROUP BY year
44  )
45  SELECT year,
46         total_revenue,
47         LAG(total_revenue) OVER (ORDER BY year) AS prev_year_revenue,
48         ROUND((total_revenue - LAG(total_revenue) OVER (ORDER BY year))
49               / LAG(total_revenue) OVER (ORDER BY year) * 100, 2) AS yoy_growth
50  FROM yearly
51  ORDER BY year;
```

Result Grid Filter Rows: Export: Wrap Cell Content:				
	year	total_revenue	prev_year_revenue	yoy_growth
►	2015	1400965638239.4607	NULL	NULL
	2016	1423573200551.3926	1400965638239.4607	1.61
	2017	1399289803363.5176	1423573200551.3926	-1.71
	2018	1407844854562.6453	1399289803363.5176	0.61
	2019	1419192469873.213	1407844854562.6453	0.81
	2020	1415826017760.0027	1419192469873.213	-0.24
	2021	1413384194408.9229	1415826017760.0027	-0.17
	2022	1394713379163.7102	1413384194408.9229	-1.32
	2023	1401175205922.6448	1394713379163.7102	0.46
	2024	1401799197092.3123	1401175205922.6448	0.04

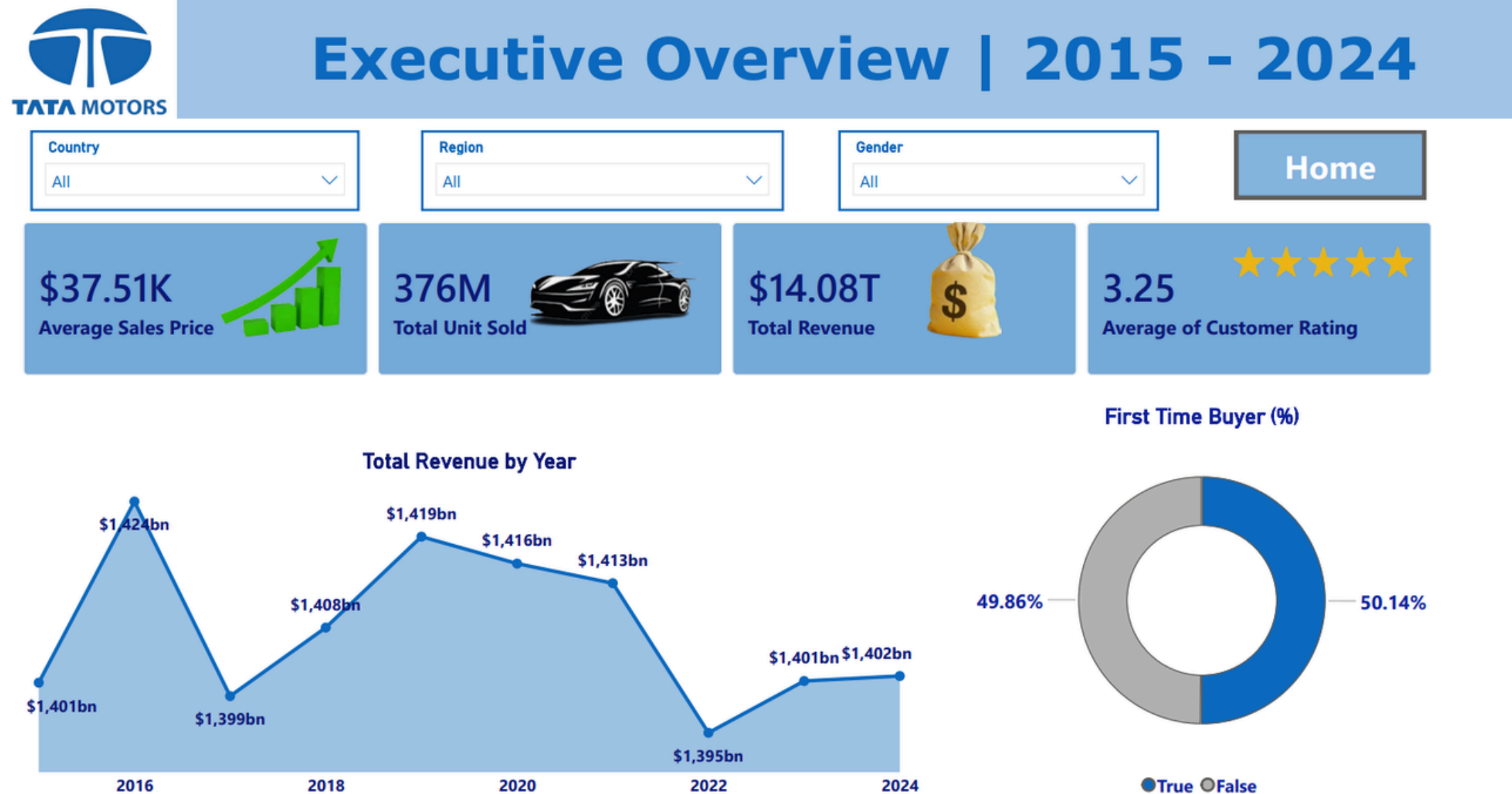
- Used LAG() Window Function
- Peak Growth: 2016 (+1.61%)
- Highest Decline: 2017 (-1.71%)
- 2022 = worst year (-1.32%)
- Insight: Revenue overall stable with minor fluctuations



Power BI Executive Dashboard

Executive Overview

This page provides a high-level summary for CEO-level decision making.



4 KPI Cards at the top show:

- Average Sale Price: \$37.51K
- Total Units Sold: 376 Million
- Total Revenue: \$14.08 Trillion
- Average Customer Rating: 3.25/5

Year-wise Revenue Trend (Line Chart):

Revenue peaked in 2016 at \$1,424bn and hit its lowest in 2022 at \$1,395bn. Overall revenue remained stable with minor fluctuations across 10 years.

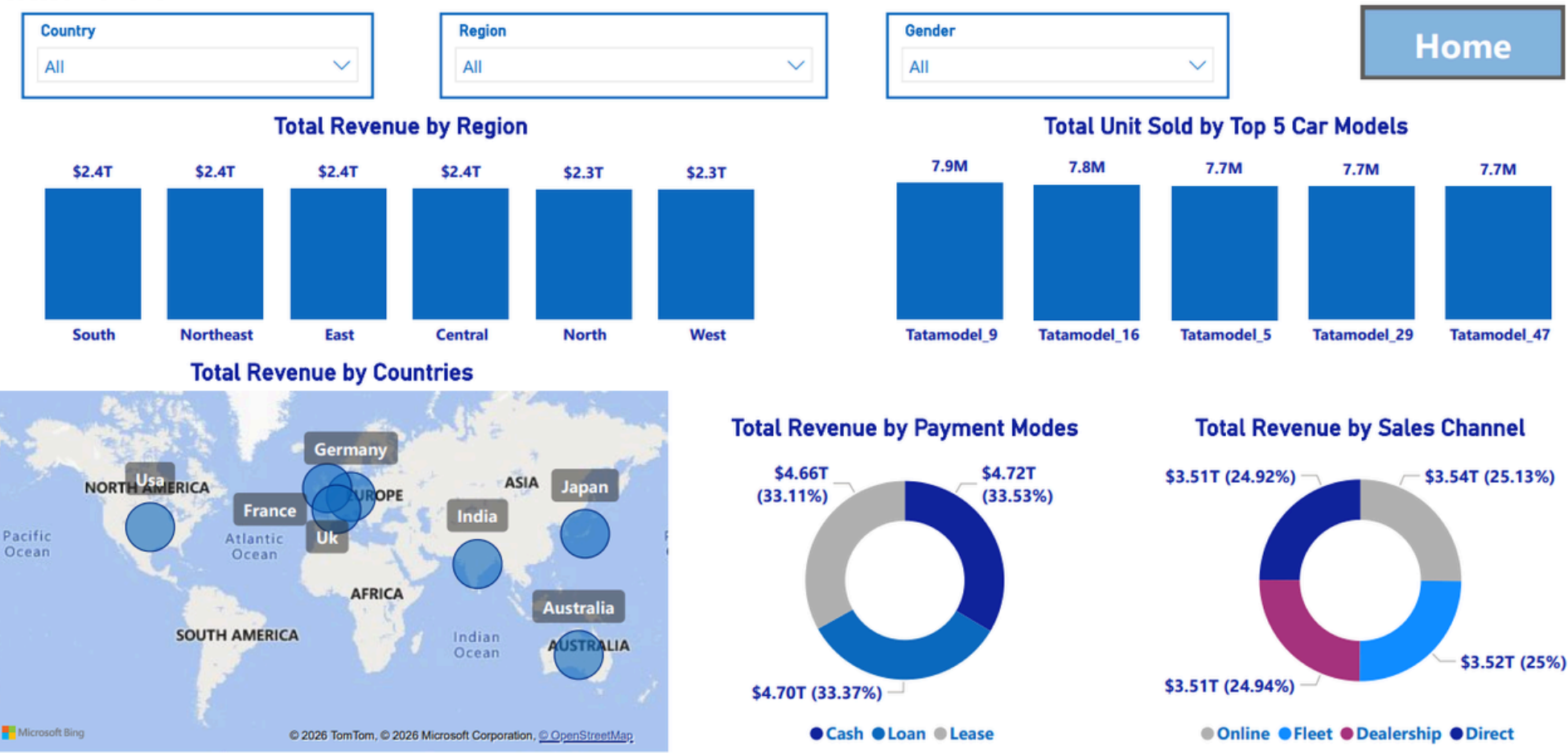
First Time Buyer Donut Chart:

50.14% are first time buyers vs 49.86% repeat buyers – almost equal split showing both new customer acquisition and retention are working equally well.

Global Sales Performance

This page gives a complete geographic and product-level view of TATA's sales.

Global Sales Performance Overview | 2015 - 2024



Total Revenue by Region

\$2.4T	\$2.4T	\$2.4T	\$2.4T	\$2.3T	\$2.3T
South	Northeast	East	Central	North	West

Total Unit Sold by Top 5 Car Models

7.9M	7.8M	7.7M	7.7M	7.7M
Tatamodel_9	Tatamodel_16	Tatamodel_5	Tatamodel_29	Tatamodel_47

Total Revenue by Countries



Total Revenue by Payment Modes

\$4.66T (33.11%)	\$4.72T (33.53%)	\$4.70T (33.37%)
Lease	Cash	Loan

Total Revenue by Sales Channel

\$3.51T (24.92%)	\$3.54T (25.13%)	\$3.52T (25%)	\$3.51T (24.94%)
Online	Fleet	Dealership	Direct

Region-wise Revenue Bar Chart:

South region leads with \$2.4T followed by Northeast and East. West region shows lowest revenue at \$2.3T. All regions are nearly equal showing uniform market penetration.

World Map Visual:

All 7 countries plotted – India, USA, UK, Germany, France, Japan, Australia showing TATA's true global presence.

Top 5 Car Models Bar Chart:

Tatamodel_9 leads with 7.9M units followed closely by Tatamodel_16 (7.8M). All top 5 models are highly competitive with minimal difference.

Payment Mode Revenue Donut:

Cash dominates at 33.53% (\$4.72T) followed by Loan (33.37%) and Lease (33.11%) – showing customers prefer outright purchase.

Sales Channel Donut:

Online channel leads at 25.13% (\$3.54T) proving TATA's digital strategy is effective.

Customer Behavior Analysis

This page reveals deep customer demographic and behavioral patterns.

Loyalty Members KPI:

50.03% customers are loyalty program members showing strong brand engagement.

Gender Distribution Donut:

Perfectly balanced – Male (33.35%), Female (33.30%), Other (33.35%) showing TATA appeals equally across all genders.

Age Group vs Income Group Matrix:

31-45 age group = highest buying segment (99.3M total units). 18-30 = lowest (86.2M). All income groups (High/Low/Medium) buy almost equally within each age group.

First Time Buyer % by Region Bar Chart:

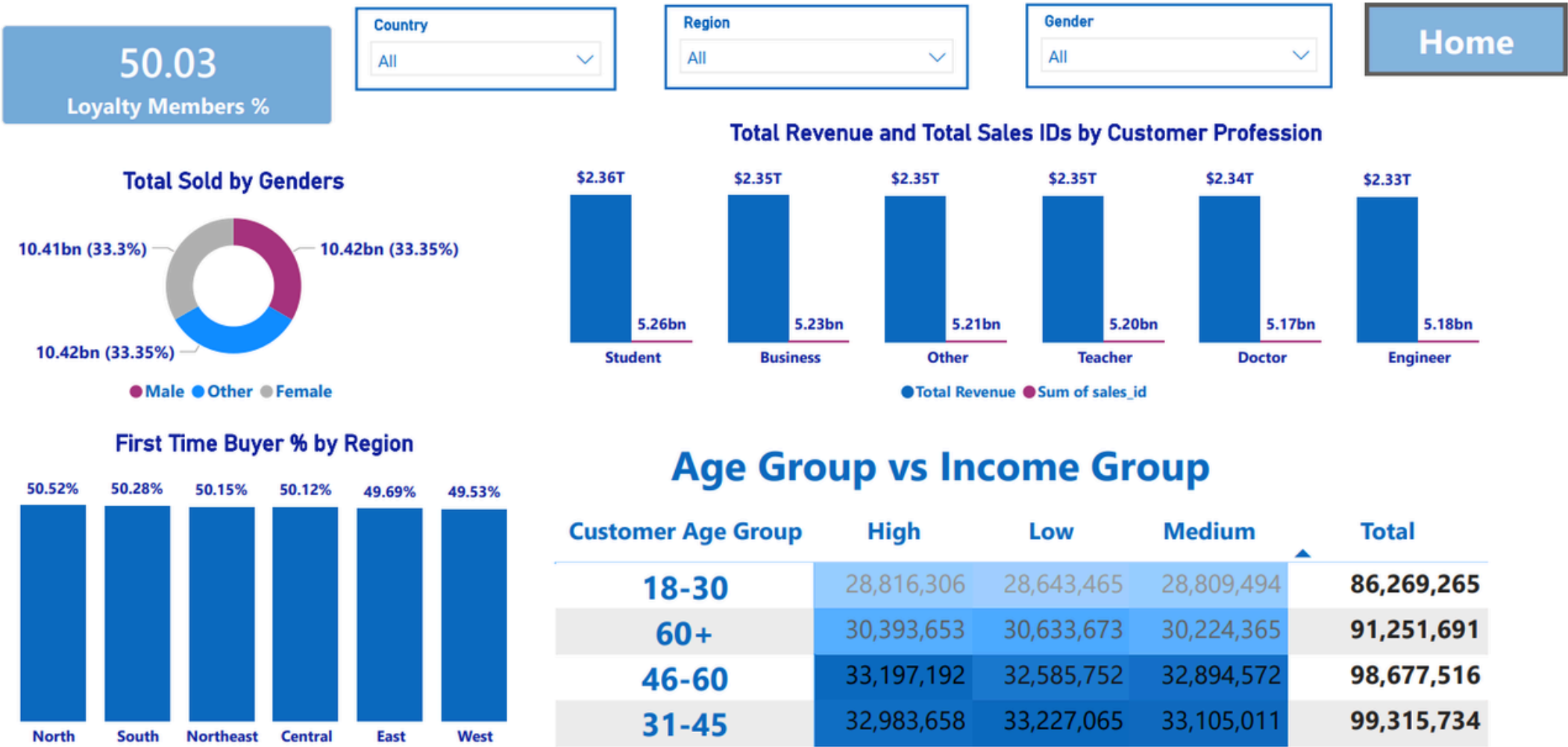
North region has highest first time buyers (50.52%) making it the strongest conquest market. West is lowest at 49.53%.

Profession-wise Revenue:

Students generate highest revenue (\$2.36T) while Engineers generate lowest (\$2.33T) – showing TATA's wide customer base across all professions.

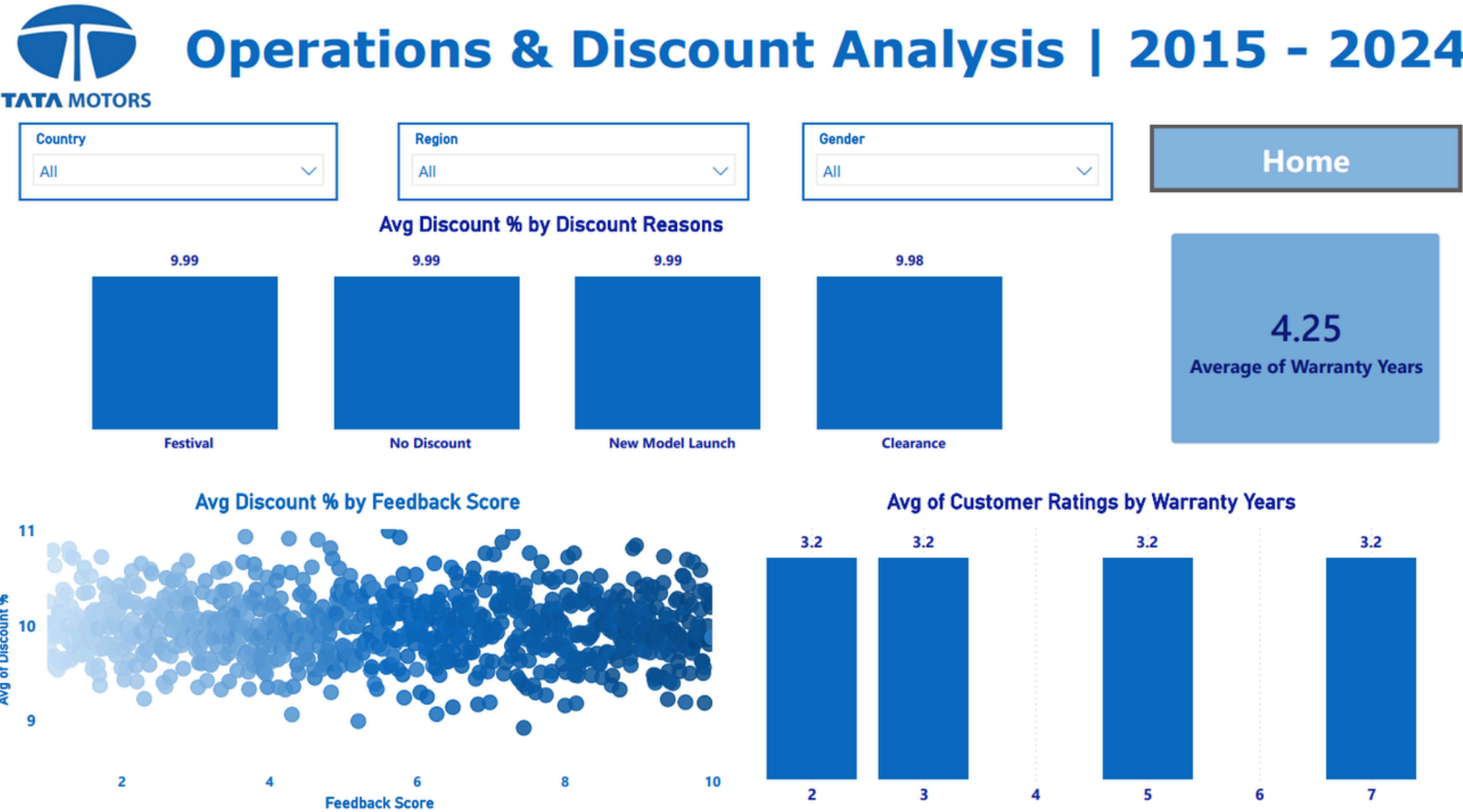


Customer Behavior Analysis | 2015 - 2024



Operations & Discount Analysis

This page focuses on financial discipline and operational efficiency.



Discount Leakage Bar Chart:

All 4 discount reasons show ~9.99% average discount. Most alarming finding – "No Discount" transactions also show 9.99% discount – clear evidence of discount leakage costing TATA significant revenue.

Discount % vs Feedback Score Scatter:

No correlation found between discount given and customer feedback score – proving that giving more discount does NOT improve customer satisfaction.

Average Warranty Years KPI: 4.25 years

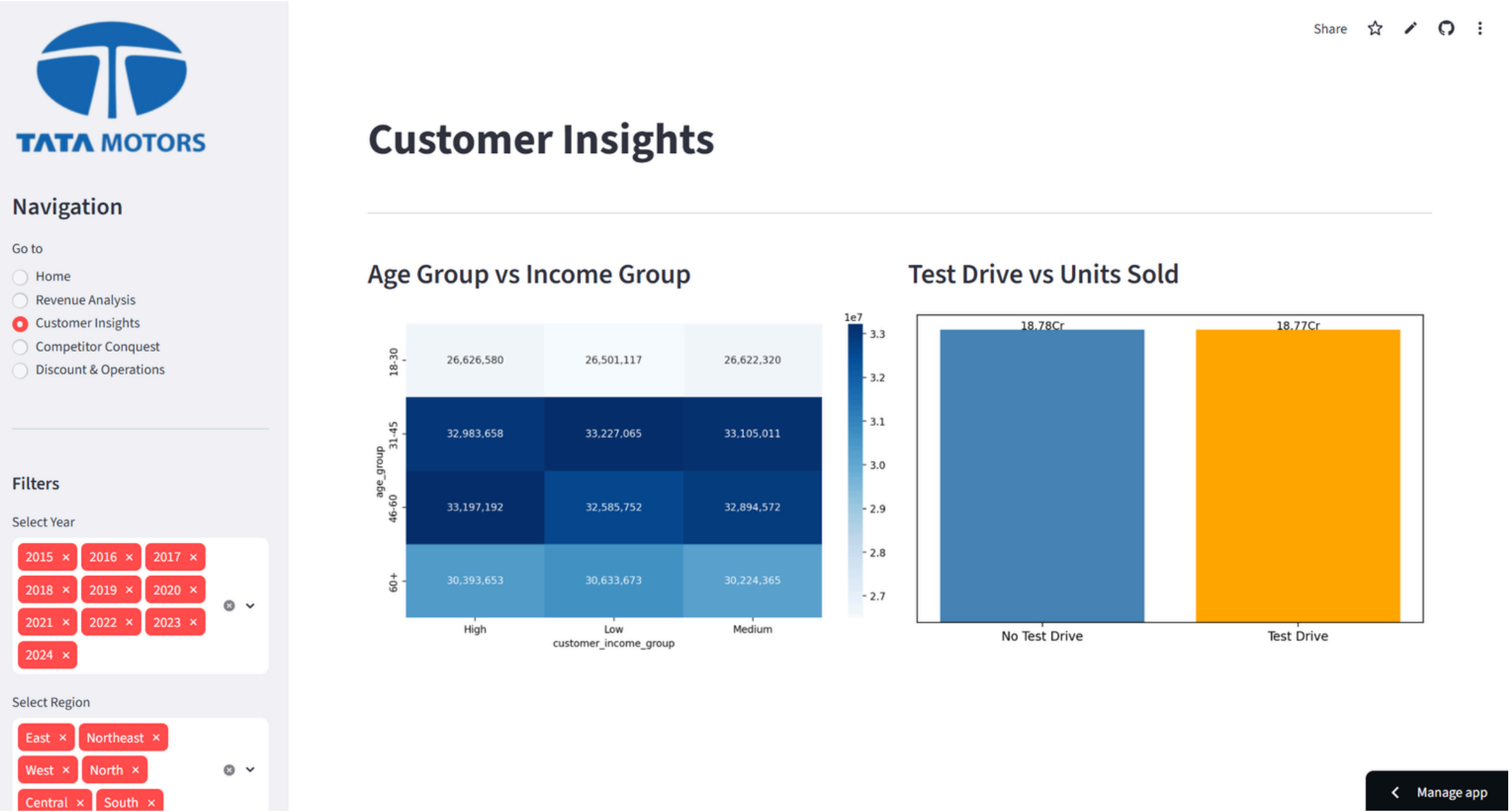
Customer Rating by Warranty Years:

All warranty durations (2-7 years) show same 3.2 rating – warranty years do not significantly impact customer satisfaction.



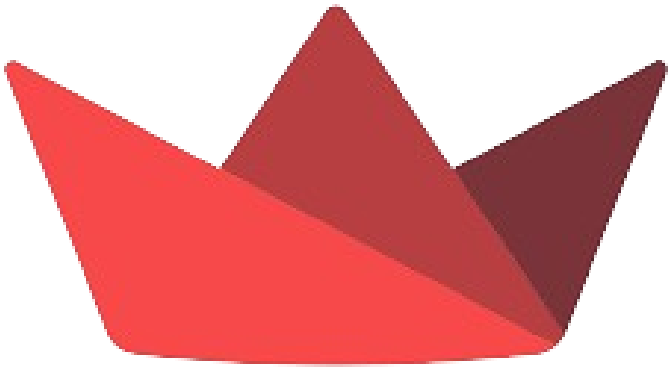
Streamlit — Live Interactive Web App

Streamlit Dashboard Screenshort



Built using Python + Streamlit

- Interactive filters — Year, Region, Country
- Home
- Revenue Analysis
- Customer Insights
- Competitor Conquest
- Discount & Operations
- Deployed on Streamlit Cloud



Live Link (Streamlit) :- [Click Here](#)

Recommendations

1. Discount Policy Reform

TATA Motors should implement strict dealer discount controls. Since "No Discount" transactions still show ~10% average discount, a real-time discount monitoring system should be introduced to prevent unnecessary revenue leakage.

01

02

2. Maruti Conquest Strategy
Since Maruti customers show lowest switching rate to TATA, a targeted marketing campaign focusing on Maruti's customer base should be launched — highlighting TATA's superior safety ratings and technology features.

3. Digital Channel Investment

Online channel contributes highest revenue at 25.13%. TATA should further invest in digital marketing, online booking experience, and virtual showrooms to maximize this growing channel.

03

04

4. Core Segment Focus

The 31-45 age group with all income levels represents TATA's strongest customer base. Dedicated loyalty programs, EMI schemes, and personalized offers for this segment will drive further revenue growth.

Conclusion

This project successfully demonstrates a complete end-to-end data analysis workflow on TATA Motors sales data comprising 250,000+ records across 7 countries and 6 regions from 2015 to 2024.

Using Python, SQL, Power BI and Streamlit, meaningful business insights were extracted covering revenue trends, customer behavior, competitor conquest and discount leakage.

Key findings reveal that TATA Motors maintains stable global revenue, with Online channel leading sales, 31-45 age group as core customers, and a critical discount leakage issue requiring immediate attention.

This analysis provides TATA Motors with actionable intelligence to optimize its sales strategy, improve customer targeting, and strengthen financial discipline across all regions and channels.

Get In Touch

- **Gmail** - *Click Here*
- **LinkedIn** - *Click Here*
- **Github Repository** - *Click Here*
- **Streamlit Dashboard Link** - *Click Here*

Thank You!

**We appreciate your valuable
time.**